# delhivery-business-case

March 10, 2026

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import ttest_ind,ttest_rel
from sklearn.preprocessing import StandardScaler, MinMaxScaler
```

**Problem Statement** Delhivery wants to analyse the data and to derive useful insights that help them advance the field as compared to their competetors. The analysis is to * clean and manipulate the data to get useful feature out of the raw data. * derive useful insights that helps the datascience team for building forecasting models * creating new features that help improving the service. * check the difference between the actual and predicted delivery metrices

## 0.1 1. Loading Dataset

[2]:
```
df=pd.read_csv('delhivery_data_business case data.txt')
```

[3]:
```
df_cpy=df.copy()
```

## 0.2 1. Basic Data Cleaning and Exploration

[ ]:
```
df.head()
```

[ ]:
```
       data          trip_creation_time  … segment_osrm_distance
segment_factor
0  training  2018-09-20 02:35:36.476840  …               11.9653
1.272727
1  training  2018-09-20 02:35:36.476840  …                9.7590
1.111111
2  training  2018-09-20 02:35:36.476840  …               10.8152
2.285714
3  training  2018-09-20 02:35:36.476840  …               13.0224
1.750000
4  training  2018-09-20 02:35:36.476840  …                3.9153
1.200000

[5 rows x 24 columns]
```

[ ]:
```
df.shape
```

[ ]: (144867, 24)

[ ]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144867 entries, 0 to 144866
Data columns (total 24 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   data                            144867 non-null  object
 1   trip_creation_time              144867 non-null  object
 2   route_schedule_uuid             144867 non-null  object
 3   route_type                      144867 non-null  object
 4   trip_uuid                       144867 non-null  object
 5   source_center                   144867 non-null  object
 6   source_name                     144574 non-null  object
 7   destination_center              144867 non-null  object
 8   destination_name                144606 non-null  object
 9   od_start_time                   144867 non-null  object
 10  od_end_time                     144867 non-null  object
 11  start_scan_to_end_scan          144867 non-null  float64
 12  is_cutoff                       144867 non-null  bool
 13  cutoff_factor                   144867 non-null  int64
 14  cutoff_timestamp                144867 non-null  object
 15  actual_distance_to_destination  144867 non-null  float64
 16  actual_time                     144867 non-null  float64
 17  osrm_time                       144867 non-null  float64
 18  osrm_distance                   144867 non-null  float64
 19  factor                          144867 non-null  float64
 20  segment_actual_time             144867 non-null  float64
 21  segment_osrm_time               144867 non-null  float64
 22  segment_osrm_distance           144867 non-null  float64
 23  segment_factor                  144867 non-null  float64
dtypes: bool(1), float64(10), int64(1), object(12)
memory usage: 25.6+ MB
```

[ ]:
```
#finding the total null values in the dataframe
df.isnull().sum()
```

[ ]:
```
data                              0
trip_creation_time                0
route_schedule_uuid               0
route_type                        0
trip_uuid                         0
source_center                     0
```

```
source_name                       293
destination_center                  0
destination_name                  261
od_start_time                       0
od_end_time                         0
start_scan_to_end_scan              0
is_cutoff                           0
cutoff_factor                       0
cutoff_timestamp                    0
actual_distance_to_destination      0
actual_time                         0
osrm_time                           0
osrm_distance                       0
factor                              0
segment_actual_time                 0
segment_osrm_time                   0
segment_osrm_distance               0
segment_factor                      0
dtype: int64
```

- There are 261 null values in destination_name column and 293 nulll values in source_name column.

[ ]:
```
# Checking for the duplicated values
df.duplicated().sum()
```

[ ]:
```
np.int64(0)
```

[ ]:
```
df.describe()
```

[ ]:
```
       start_scan_to_end_scan  …  segment_factor
count           144867.000000  …   144867.000000
mean               961.262986  …        2.218368
std               1037.012769  …        4.847530
min                 20.000000  …      -23.444444
25%                161.000000  …        1.347826
50%                449.000000  …        1.684211
75%               1634.000000  …        2.250000
max               7898.000000  …      574.250000

[8 rows x 11 columns]
```

[4]:
```
# Converting into the pd datetime format
df['trip_creation_time']=pd.to_datetime(df['trip_creation_time'])
df['od_start_time'] = pd.to_datetime(df['od_start_time'])
df['od_end_time'] = pd.to_datetime(df['od_end_time'])
```

## 0.3 2. Exploratory Data Analysis

[ ]:
```
df['data'].value_counts()
```

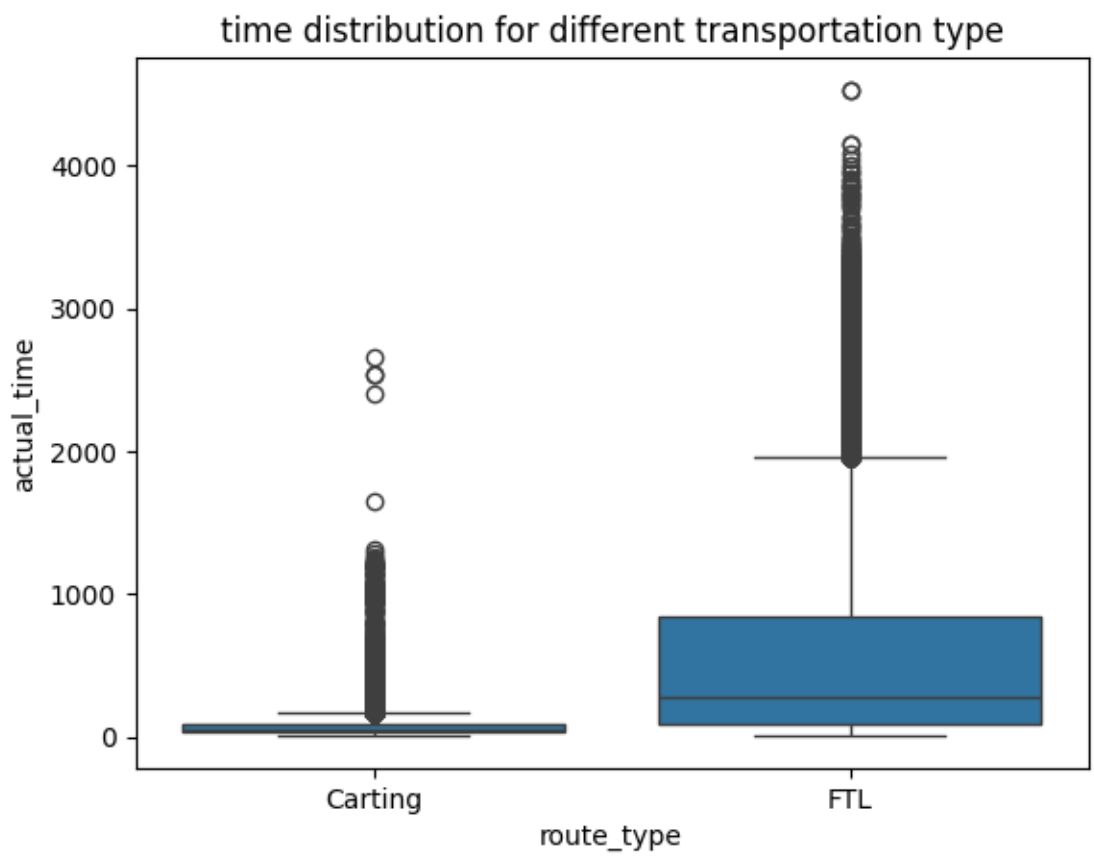
[ ]:
```
data
training    104858
test         40009
Name: count, dtype: int64
```

[ ]:
```
df['route_type'].value_counts()
```

[ ]:
```
route_type
FTL        99660
Carting    45207
Name: count, dtype: int64
```

[ ]:
```
sns.boxplot(x='route_type',y='actual_time',data=df)
plt.title('time distribution for different transportation type')
```

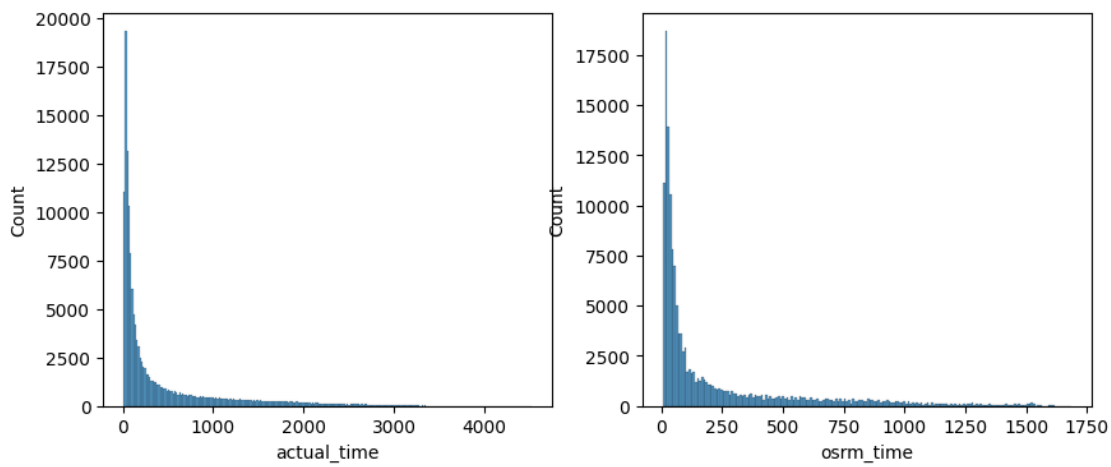
[ ]:
```
Text(0.5, 1.0, 'time distribution for different transportation type')
```



**Insight** : FTL is more preferred transportation as the total time is lesser there

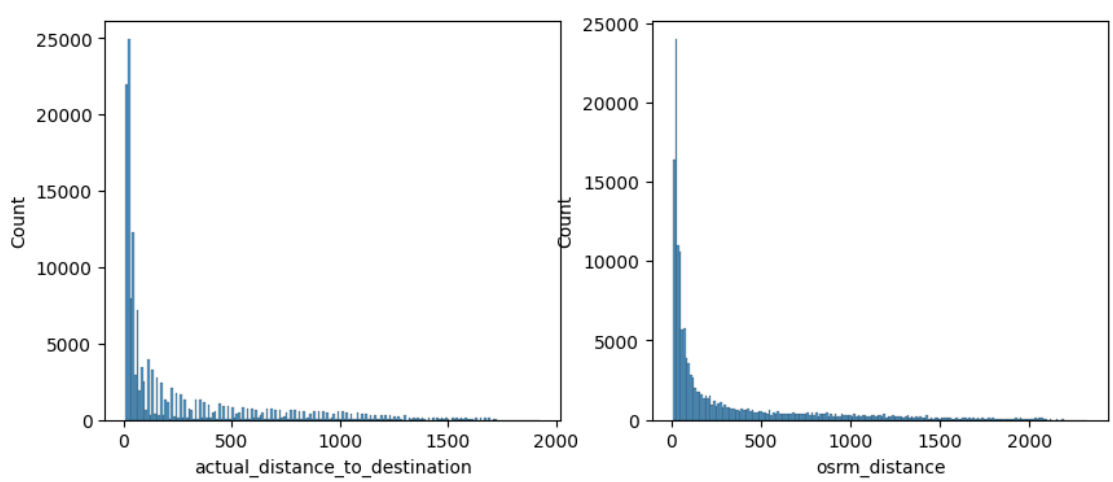
[ ]:
```
plt.figure(figsize=(10,4)),
plt.subplot(1,2,1)
sns.histplot(df['actual_time'])
plt.subplot(1,2,2)
sns.histplot(df['osrm_time'])
```

[ ]: <Axes: xlabel='osrm_time', ylabel='Count'>



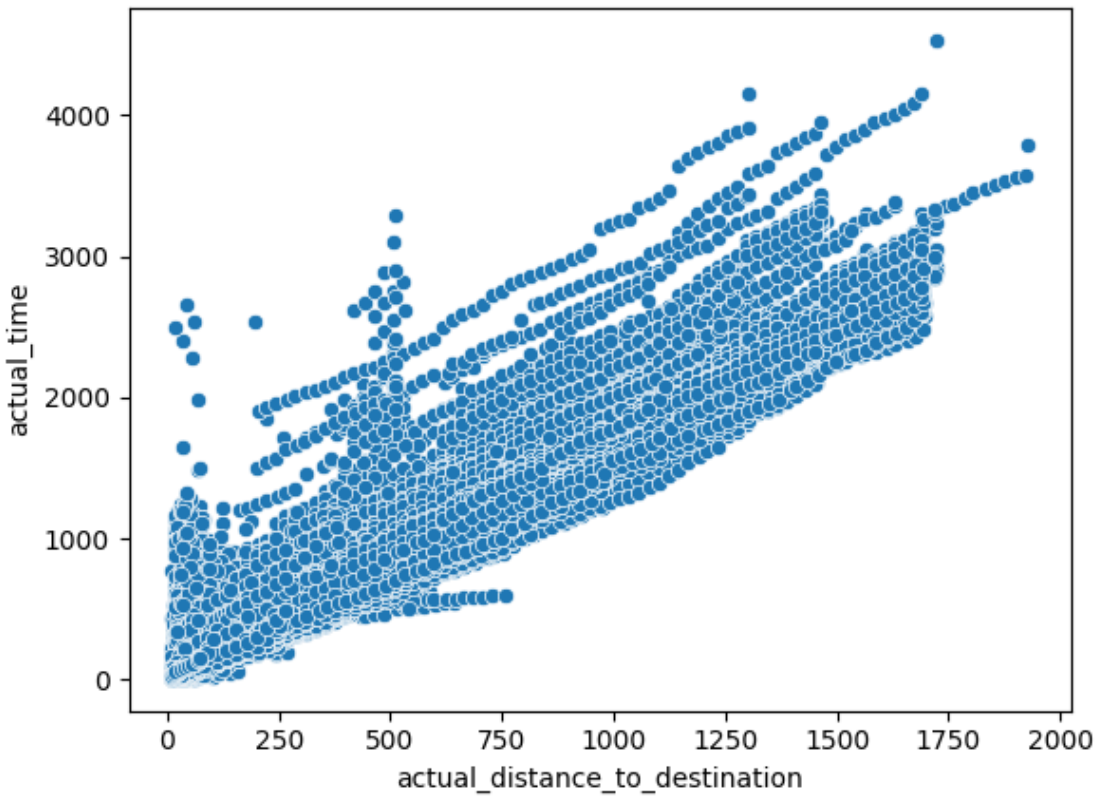

[ ]:
```
plt.figure(figsize=(10,4)),
plt.subplot(1,2,1)
sns.histplot(df['actual_distance_to_destination'])
plt.subplot(1,2,2)
sns.histplot(df['osrm_distance'])
```

[ ]: <Axes: xlabel='osrm_distance', ylabel='Count'>



[44]:
```
sns.
↪scatterplot(x=df['actual_distance_to_destination'],y=df['actual_time'],data=df)
```

[44]: <Axes: xlabel='actual_distance_to_destination', ylabel='actual_time'>




- As the distance increases time also increases
- A few deviations are also visible in the linear pattern

## 0.4 2. Merging Rows

[5]:
```
# creating a new column segment_key by merging trip_uuid, source_center,␣
↪destination_center
df['segment_key']=df['trip_uuid']+'_'+df['source_center']+'_'+df['destination_center']
df['segment_key']
```

[5]:
```
0          trip-153741093647649320_IND388121AAA_IND388620AAB
1          trip-153741093647649320_IND388121AAA_IND388620AAB
2          trip-153741093647649320_IND388121AAA_IND388620AAB
3          trip-153741093647649320_IND388121AAA_IND388620AAB
```

```
4         trip-153741093647649320_IND388121AAA_IND388620AAB
                                   …
144862    trip-153746066843555182_IND131028AAB_IND000000ACB
144863    trip-153746066843555182_IND131028AAB_IND000000ACB
144864    trip-153746066843555182_IND131028AAB_IND000000ACB
144865    trip-153746066843555182_IND131028AAB_IND000000ACB
144866    trip-153746066843555182_IND131028AAB_IND000000ACB
Name: segment_key, Length: 144867, dtype: object
```

[6]:
```
# Grouping on the numerical columns by the segment_key
df['segment_actual_time_sum']=df.groupby('segment_key')['actual_time'].cumsum()
df['segment_osrm_time_sum']=df.groupby('segment_key')['osrm_time'].cumsum()
df['segment_osrm_distance_sum']=df.groupby('segment_key')['osrm_distance'].
 ↪cumsum()
df[['segment_actual_time_sum','segment_osrm_time_sum','segment_osrm_distance_sum']]
```

[6]:
```
        segment_actual_time_sum  segment_osrm_time_sum  \
0                          14.0                   11.0
1                          38.0                   31.0
2                          78.0                   59.0
3                         140.0                   99.0
4                         208.0                  143.0
…                           …                      …
144862                    345.0                  230.0
144863                    465.0                  306.0
144864                    605.0                  394.0
144865                    763.0                  492.0
144866                   1189.0                  587.0

        segment_osrm_distance_sum
0                         11.9653
1                         33.6896
2                         66.2291
3                        111.7911
4                        166.0092
…                           …
144862                   228.5453
144863                   314.2282
144864                   411.3215
144865                   522.5924
144866                   611.3243

[144867 rows x 3 columns]
```

[7]:
```
# Creating dictionary for aggreation on numerical columns
```

```
create_segment_dict={'trip_uuid':'first','od_start_time':'first','od_end_time':
 ↪'last','actual_time':'sum','osrm_time':'sum','osrm_distance':
 ↪'sum','segment_actual_time':'sum','segment_osrm_time':
 ↪'sum','segment_osrm_distance':'sum'}
```

[8]:
```
# Aggregation based on the segment_key
segment_base_agg=df.groupby('segment_key').agg(create_segment_dict).
 ↪sort_values('segment_key',ascending=True)
segment_base_agg.sort_values('od_end_time',ascending=True)
```

[8]:
```
                                                                  trip_uuid  \
segment_key
trip-153671110078355292_IND121004AAB_IND121001AAA  trip-153671110078355292
trip-153671079956500691_IND110024AAA_IND110014AAA  trip-153671079956500691
trip-153671066826362165_IND560043AAC_IND560064AAA  trip-153671066826362165
trip-153671173668736946_IND110043AAA_IND110078AAA  trip-153671173668736946
trip-153671277074687197_IND624001AAA_IND624619AAA  trip-153671277074687197
…                                                                        …
trip-153859003271955591_IND000000ACB_IND562132AAA  trip-153859003271955591
trip-153840656812932039_IND712311AAA_IND421302AAG  trip-153840656812932039
trip-153858876340944305_IND000000ACB_IND562132AAA  trip-153858876340944305
trip-153860879439383883_IND000000ACB_IND160002AAC  trip-153860879439383883
trip-153843695443252828_IND764071AAB_IND530012AAA  trip-153843695443252828

                                                                od_start_time  \
segment_key
trip-153671110078355292_IND121004AAB_IND121001AAA 2018-09-12 00:11:40.783923
trip-153671079956500691_IND110024AAA_IND110014AAA 2018-09-12 00:06:39.565253
trip-153671066826362165_IND560043AAC_IND560064AAA 2018-09-12 00:04:28.263977
trip-153671173668736946_IND110043AAA_IND110078AAA 2018-09-12 00:22:16.687619
trip-153671277074687197_IND624001AAA_IND624619AAA 2018-09-12 00:39:30.747127
…                                                                        …
trip-153859003271955591_IND000000ACB_IND562132AAA 2018-10-04 05:27:22.883248
trip-153840656812932039_IND712311AAA_IND421302AAG 2018-10-03 21:21:11.054680
trip-153858876340944305_IND000000ACB_IND562132AAA 2018-10-03 17:46:03.409692
trip-153860879439383883_IND000000ACB_IND160002AAC 2018-10-06 04:27:23.392375
trip-153843695443252828_IND764071AAB_IND530012AAA 2018-10-02 15:21:51.236205

                                                                  od_end_time  \
segment_key
trip-153671110078355292_IND121004AAB_IND121001AAA 2018-09-12 00:50:10.814399
trip-153671079956500691_IND110024AAA_IND110014AAA 2018-09-12 00:55:59.568645
trip-153671066826362165_IND560043AAC_IND560064AAA 2018-09-12 01:13:03.560690
trip-153671173668736946_IND110043AAA_IND110078AAA 2018-09-12 01:29:19.277412
trip-153671277074687197_IND624001AAA_IND624619AAA 2018-09-12 01:32:05.649177
…                                                                        …
trip-153859003271955591_IND000000ACB_IND562132AAA 2018-10-06 11:08:18.498393
```

```
trip-153840656812932039_IND712311AAA_IND421302AAG 2018-10-06 19:28:24.500705
trip-153858876340944305_IND000000ACB_IND562132AAA 2018-10-06 19:46:59.723686
trip-153860879439383883_IND000000ACB_IND160002AAC 2018-10-07 00:50:44.569308
trip-153843695443252828_IND764071AAB_IND530012AAA 2018-10-08 03:00:24.353479

                                                   actual_time  osrm_time  \
segment_key
trip-153671110078355292_IND121004AAB_IND121001AAA         17.0        9.0
trip-153671079956500691_IND110024AAA_IND110014AAA         35.0       16.0
trip-153671066826362165_IND560043AAC_IND560064AAA         50.0       36.0
trip-153671173668736946_IND110043AAA_IND110078AAA         72.0       42.0
trip-153671277074687197_IND624001AAA_IND624619AAA         64.0       48.0
…                                                          …        …
trip-153859003271955591_IND000000ACB_IND562132AAA     107774.0    60709.0
trip-153840656812932039_IND712311AAA_IND421302AAG     120992.0    51677.0
trip-153858876340944305_IND000000ACB_IND562132AAA     167920.0    60504.0
trip-153860879439383883_IND000000ACB_IND160002AAC       2626.0     1439.0
trip-153843695443252828_IND764071AAB_IND530012AAA       2541.0      211.0

                                                   osrm_distance  \
segment_key
trip-153671110078355292_IND121004AAB_IND121001AAA        10.8159
trip-153671079956500691_IND110024AAA_IND110014AAA        19.9606
trip-153671066826362165_IND560043AAC_IND560064AAA        27.6866
trip-153671173668736946_IND110043AAA_IND110078AAA        37.2326
trip-153671277074687197_IND624001AAA_IND624619AAA        51.3800
…                                                          …
trip-153859003271955591_IND000000ACB_IND562132AAA     79759.7761
trip-153840656812932039_IND712311AAA_IND421302AAG     72491.2403
trip-153858876340944305_IND000000ACB_IND562132AAA     79458.9058
trip-153860879439383883_IND000000ACB_IND160002AAC      1997.8380
trip-153843695443252828_IND764071AAB_IND530012AAA       293.3271

                                                   segment_actual_time  \
segment_key
trip-153671110078355292_IND121004AAB_IND121001AAA                 17.0
trip-153671079956500691_IND110024AAA_IND110014AAA                 23.0
trip-153671066826362165_IND560043AAC_IND560064AAA                 25.0
trip-153671173668736946_IND110043AAA_IND110078AAA                 43.0
trip-153671277074687197_IND624001AAA_IND624619AAA                 40.0
…                                                                  …
trip-153859003271955591_IND000000ACB_IND562132AAA               2917.0
trip-153840656812932039_IND712311AAA_IND421302AAG               3338.0
trip-153858876340944305_IND000000ACB_IND562132AAA               4122.0
trip-153860879439383883_IND000000ACB_IND160002AAC                438.0
trip-153843695443252828_IND764071AAB_IND530012AAA               2541.0
```

```
                                                      segment_osrm_time  \
segment_key
trip-153671110078355292_IND121004AAB_IND121001AAA                   9.0
trip-153671079956500691_IND110024AAA_IND110014AAA                  14.0
trip-153671066826362165_IND560043AAC_IND560064AAA                  18.0
trip-153671173668736946_IND110043AAA_IND110078AAA                  25.0
trip-153671277074687197_IND624001AAA_IND624619AAA                  26.0
…                                                                   …
trip-153859003271955591_IND000000ACB_IND562132AAA                1860.0
trip-153840656812932039_IND712311AAA_IND421302AAG                1431.0
trip-153858876340944305_IND000000ACB_IND562132AAA                1706.0
trip-153860879439383883_IND000000ACB_IND160002AAC                 232.0
trip-153843695443252828_IND764071AAB_IND530012AAA                 211.0

                                                      segment_osrm_distance
segment_key
trip-153671110078355292_IND121004AAB_IND121001AAA                   10.8159
trip-153671079956500691_IND110024AAA_IND110014AAA                   16.0860
trip-153671066826362165_IND560043AAC_IND560064AAA                   13.8433
trip-153671173668736946_IND110043AAA_IND110078AAA                   22.6548
trip-153671277074687197_IND624001AAA_IND624619AAA                   27.8572
…                                                                      …
trip-153859003271955591_IND000000ACB_IND562132AAA                 2424.6878
trip-153840656812932039_IND712311AAA_IND421302AAG                 2068.1202
trip-153858876340944305_IND000000ACB_IND562132AAA                 2242.9029
trip-153860879439383883_IND000000ACB_IND160002AAC                  312.8295
trip-153843695443252828_IND764071AAB_IND530012AAA                  293.3271

[26368 rows x 9 columns]
```

## 0.5 3.Feature Engineering

[9]:
```python
# Creating new feature 'od_time_diff_hour' from 'od_start_time' and␣
↪'od_end_time'
df['od_time_diff_hour']=(df['od_end_time']-df['od_start_time']).dt.
↪total_seconds()/3600
df.drop(['od_end_time','od_start_time'],axis=1,inplace=True)
df.head()
```

[9]:
```
        data         trip_creation_time  \
0  training 2018-09-20 02:35:36.476840
1  training 2018-09-20 02:35:36.476840
2  training 2018-09-20 02:35:36.476840
3  training 2018-09-20 02:35:36.476840
4  training 2018-09-20 02:35:36.476840
```

```
                                route_schedule_uuid route_type  \
0  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
1  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
2  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
3  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
4  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting

                trip_uuid source_center                 source_name  \
0  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
1  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
2  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
3  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
4  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)

  destination_center               destination_name  start_scan_to_end_scan  \
0       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
1       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
2       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
3       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
4       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0

   …    factor  segment_actual_time segment_osrm_time  \
0  …  1.272727                 14.0              11.0
1  …  1.200000                 10.0               9.0
2  …  1.428571                 16.0               7.0
3  …  1.550000                 21.0              12.0
4  …  1.545455                  6.0               5.0

   segment_osrm_distance  segment_factor  \
0                11.9653        1.272727
1                 9.7590        1.111111
2                10.8152        2.285714
3                13.0224        1.750000
4                 3.9153        1.200000

                                        segment_key  segment_actual_time_sum  \
0  trip-153741093647649320_IND388121AAA_IND388620AAB                     14.0
1  trip-153741093647649320_IND388121AAA_IND388620AAB                     38.0
2  trip-153741093647649320_IND388121AAA_IND388620AAB                     78.0
3  trip-153741093647649320_IND388121AAA_IND388620AAB                    140.0
4  trip-153741093647649320_IND388121AAA_IND388620AAB                    208.0

   segment_osrm_time_sum  segment_osrm_distance_sum  od_time_diff_hour
0                   11.0                    11.9653           1.436894
1                   31.0                    33.6896           1.436894
2                   59.0                    66.2291           1.436894
3                   99.0                   111.7911           1.436894
```

```
4                  143.0                    166.0092           1.436894

[5 rows x 27 columns]
```

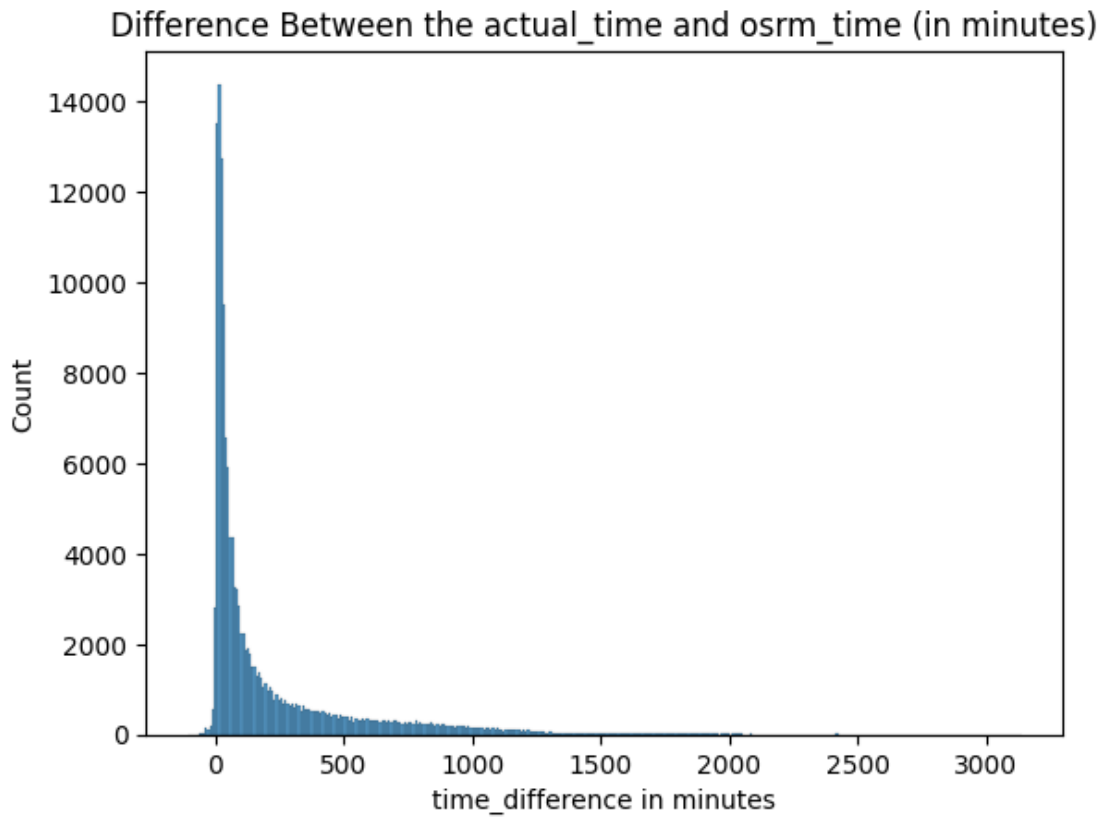
[70]:
```python
# time difference between the actual time taken and the predicted time for each␣
↪trip
df['tim_diff'] = df['actual_time'] - df['osrm_time']
df['tim_diff'].describe()
```

[70]:
```
count    144867.000000
mean        203.059254
std         303.743664
min        -110.000000
25%          21.000000
50%          65.000000
75%         247.000000
max        3137.000000
Name: tim_diff, dtype: float64
```

- Mean time difference between the actual time and predicted time is 203.
- It tells that the actual time taken is more than the predicted time quite often

[76]:
```python
sns.histplot(df['tim_diff'])
plt.title('Difference Between the actual_time and osrm_time (in minutes)')
plt.xlabel('time_difference in minutes')
```

[76]:
```
Text(0.5, 0, 'time_difference in minutes')
```




- There is difference between the actual time and predicted time. Only a few deliveries are happening before time

[11]:
```
# Splitting and extracting features out of the source_name
df[['source_city','source_place','source_code']]=df['source_name'].str.
 ↪split('_',n=2,expand=True)
df.head()
```

[11]:
```
        data         trip_creation_time  \
0  training 2018-09-20 02:35:36.476840
1  training 2018-09-20 02:35:36.476840
2  training 2018-09-20 02:35:36.476840
3  training 2018-09-20 02:35:36.476840
4  training 2018-09-20 02:35:36.476840

                                route_schedule_uuid route_type  \
0  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
1  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
2  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
3  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
4  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
```

```
                 trip_uuid source_center                 source_name  \
0  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
1  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
2  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
3  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
4  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)

  destination_center               destination_name  start_scan_to_end_scan  \
0       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
1       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
2       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
3       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
4       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0

   …  segment_factor                                        segment_key  \
0  …        1.272727  trip-153741093647649320_IND388121AAA_IND388620AAB
1  …        1.111111  trip-153741093647649320_IND388121AAA_IND388620AAB
2  …        2.285714  trip-153741093647649320_IND388121AAA_IND388620AAB
3  …        1.750000  trip-153741093647649320_IND388121AAA_IND388620AAB
4  …        1.200000  trip-153741093647649320_IND388121AAA_IND388620AAB

  segment_actual_time_sum  segment_osrm_time_sum  segment_osrm_distance_sum  \
0                    14.0                   11.0                    11.9653
1                    38.0                   31.0                    33.6896
2                    78.0                   59.0                    66.2291
3                   140.0                   99.0                   111.7911
4                   208.0                  143.0                   166.0092

   od_time_diff_hour  tim_diff  source_city  source_place   source_code
0           1.436894       3.0        Anand       VUNagar  DC (Gujarat)
1           1.436894       4.0        Anand       VUNagar  DC (Gujarat)
2           1.436894      12.0        Anand       VUNagar  DC (Gujarat)
3           1.436894      22.0        Anand       VUNagar  DC (Gujarat)
4           1.436894      24.0        Anand       VUNagar  DC (Gujarat)

[5 rows x 31 columns]
```

[12]:
```
# finding the state where the most number of orders are coming from
source_statewise_order=df.groupby('source_code')['trip_uuid'].count().
 ↪sort_values(ascending=False)
source_statewise_order.head()
```

[12]:
```
source_code
HB (Haryana)         23347
H (Karnataka)        10061
HB (Maharashtra)      9088
```

```
H (Maharashtra)       4064
D (Uttar Pradesh)     3415
Name: trip_uuid, dtype: int64
```

- Most number of orders are coming from Haryana.
- Nearly considerable number of order are shipped from boht Karnataka and Maharashtra

[13]:
```
# Extracting features out of the destination_name
df[['destination_city','destination_place','destination_code']]=df['destination_name'].
 ↪str.split('_',n=2,expand=True)
df.head()
```

[13]:
```
       data         trip_creation_time  \
0  training 2018-09-20 02:35:36.476840
1  training 2018-09-20 02:35:36.476840
2  training 2018-09-20 02:35:36.476840
3  training 2018-09-20 02:35:36.476840
4  training 2018-09-20 02:35:36.476840

                                route_schedule_uuid route_type  \
0  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
1  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
2  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
3  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
4  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting

                 trip_uuid source_center                 source_name  \
0  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
1  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
2  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
3  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
4  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)

  destination_center               destination_name  start_scan_to_end_scan  \
0       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
1       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
2       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
3       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
4       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0

   …  segment_osrm_time_sum  segment_osrm_distance_sum od_time_diff_hour  \
0  …                   11.0                    11.9653          1.436894
1  …                   31.0                    33.6896          1.436894
2  …                   59.0                    66.2291          1.436894
3  …                   99.0                   111.7911          1.436894
4  …                  143.0                   166.0092          1.436894
```

```
   tim_diff  source_city  source_place   source_code  destination_city  \
0       3.0        Anand       VUNagar  DC (Gujarat)          Khambhat
1       4.0        Anand       VUNagar  DC (Gujarat)          Khambhat
2      12.0        Anand       VUNagar  DC (Gujarat)          Khambhat
3      22.0        Anand       VUNagar  DC (Gujarat)          Khambhat
4      24.0        Anand       VUNagar  DC (Gujarat)          Khambhat

   destination_place  destination_code
0           MotvdDPP       D (Gujarat)
1           MotvdDPP       D (Gujarat)
2           MotvdDPP       D (Gujarat)
3           MotvdDPP       D (Gujarat)
4           MotvdDPP       D (Gujarat)

[5 rows x 34 columns]
```

[14]:
```
# finding the destination wise orders, where to the maximum number of orders␣
↪are delivered
destination_statewisea_order=df.groupby('destination_code')['trip_uuid'].
↪count().sort_values(ascending=False)
destination_statewisea_order.head()
```

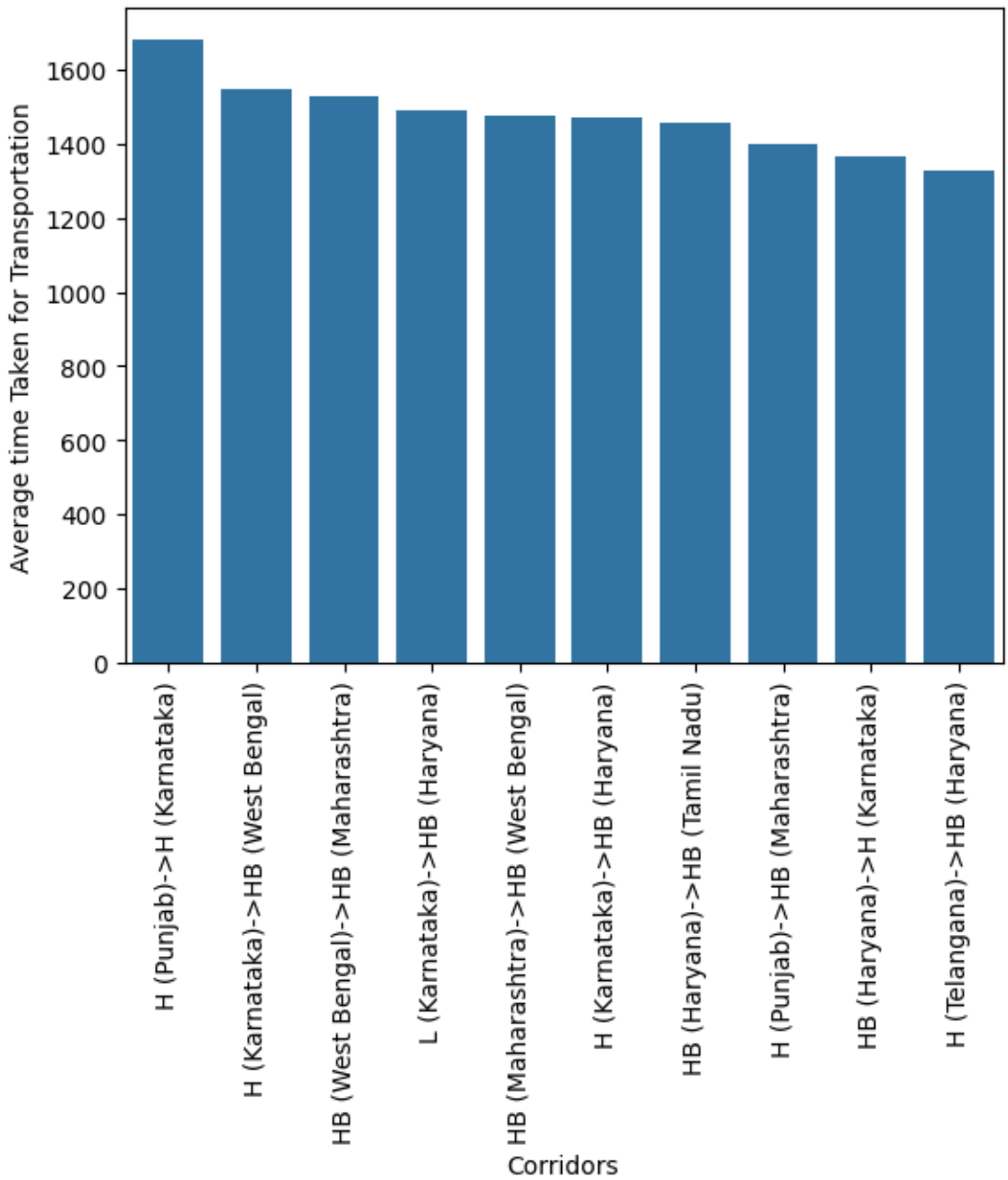
[14]:
```
destination_code
HB (Haryana)        15192
H (Karnataka)       11274
HB (Maharashtra)     5492
H (Telangana)        5142
HB (West Bengal)     4892
Name: trip_uuid, dtype: int64
```

- Most number of orders are also delivered to Haryana and then Karnataka

[15]:
```
# creating a new feature corridor with source code and destination code
df['corridor'] = df['source_code'] + '->' + df['destination_code']
```

[65]:
```
corridorwise_time=df.groupby('corridor')['actual_time'].mean().
↪sort_values(ascending=False)
sns.barplot(x=corridorwise_time.head(10).index,y=corridorwise_time.head(10).
↪values)
plt.xticks(rotation=90)
plt.xlabel('Corridors')
plt.ylabel('Average time Taken for Transportation')
```

[65]:
```
Text(0, 0.5, 'Average time Taken for Transportation')
```




- Average time for tranportation is more for the corridor Punjab to Karnataka

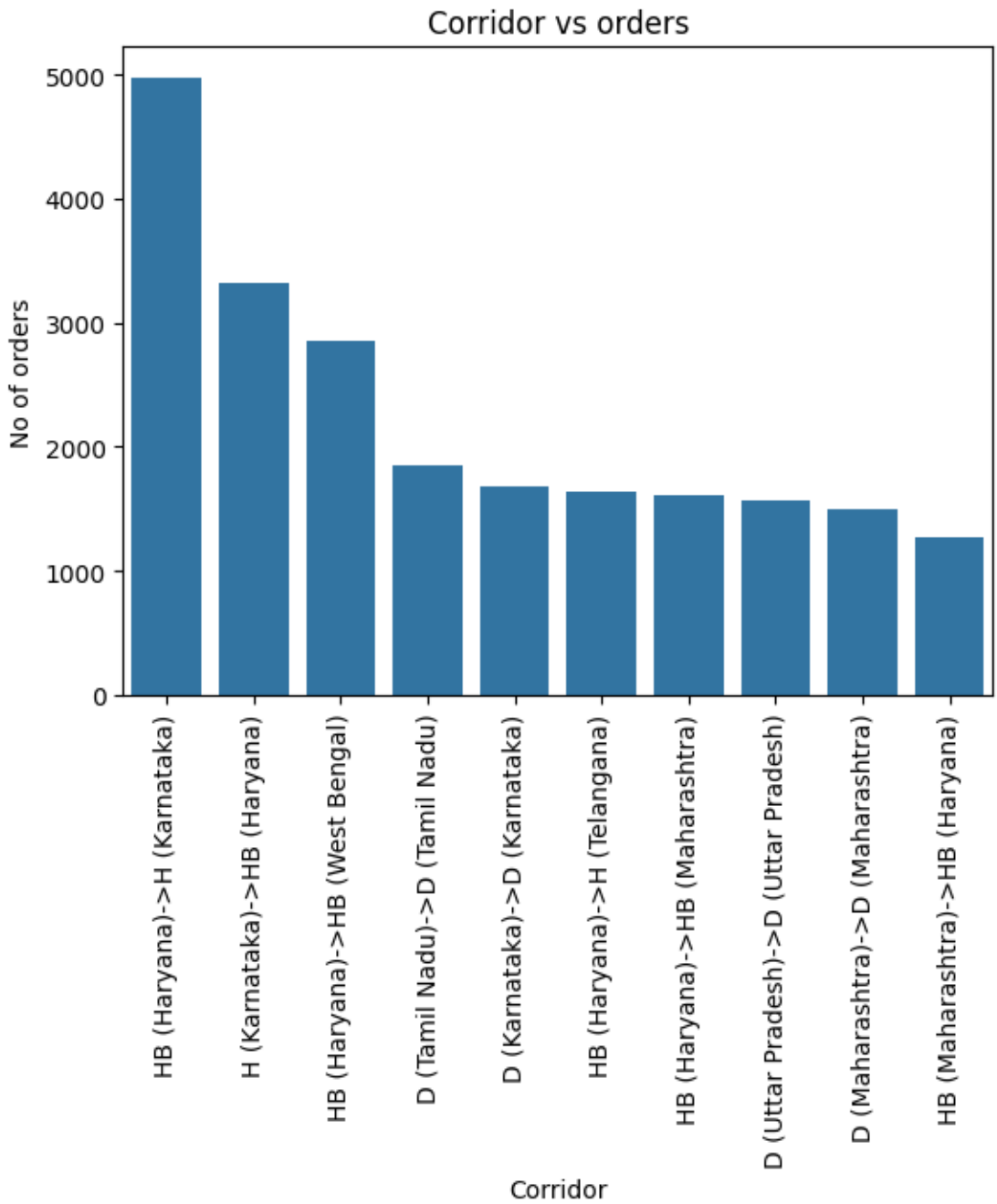
[16]:
```
# finding the busiest corridor
corridorwise_orders = df.groupby('corridor')['trip_uuid'].count().
 ↪sort_values(ascending=False)
corridorwise_orders.head()
```

[16]:
```
corridor
HB (Haryana)->H (Karnataka)       4976
H (Karnataka)->HB (Haryana)       3316
HB (Haryana)->HB (West Bengal)    2862
D (Tamil Nadu)->D (Tamil Nadu)    1846
D (Karnataka)->D (Karnataka)      1682
Name: trip_uuid, dtype: int64
```

- Largest number of transportation happens from Haryana to Karnataka, seems to be the busiest one.
- Karnataka to Haryana is the next busiest one.

[58]:
```
sns.barplot(x=corridorwise_orders.head(10).index,y=corridorwise_orders.head(10).
 ↪values)
plt.xticks(rotation=90)
plt.title('Corridor vs orders')
plt.xlabel('Corridor')
plt.ylabel('No of orders')
```

[58]:
```
Text(0, 0.5, 'No of orders')
```




[17]:
```
df['city_to_city'] = df['source_city'] + '->' + df['destination_city']
```

[18]:
```
city_to_city_routes = df.groupby('city_to_city')['trip_uuid'].agg('count')
```

[ ]:
```
city_to_city_routes.head()
```

[ ]:
```
city_to_city
Gurgaon->Bangalore      4976
Bangalore->Gurgaon      3316
Gurgaon->Kolkata        2862
Bengaluru->Bengaluru    2062
Bangalore->Bengaluru    1741
Name: trip_uuid, dtype: int64
```

- Busiest transportation route between cities is from Guragon to Banglore, then backwards

[19]:
```
# distance for different corridors
source_to_destination_distance=df.
 ↪groupby('corridor')['actual_distance_to_destination'].mean().
 ↪sort_values(ascending=False)
source_to_destination_distance.head()
```

[19]:
```
corridor
H (Punjab)->H (Karnataka)        1050.751668
HB (Haryana)->HB (Tamil Nadu)     876.628008
L (Karnataka)->HB (Haryana)       870.118858
H (Karnataka)->HB (Haryana)       869.072245
HB (Haryana)->H (Karnataka)       859.827666
Name: actual_distance_to_destination, dtype: float64
```

- Punjab to Karnataka is the longest transporation route

[20]:
```
# Finding the time for each corridor
source_to_destination_time=df.groupby('corridor')['actual_time'].mean().
 ↪sort_values(ascending=False)
source_to_destination_time.head()
```

[20]:
```
corridor
H (Punjab)->H (Karnataka)             1682.728395
H (Karnataka)->HB (West Bengal)       1549.171296
HB (West Bengal)->HB (Maharashtra)    1531.104278
L (Karnataka)->HB (Haryana)           1489.641026
HB (Maharashtra)->HB (West Bengal)    1477.367316
Name: actual_time, dtype: float64
```

[21]:
```
# Extracting features out of the trip_creation_time
df['trip_creation_year'] = df['trip_creation_time'].dt.year
df['trip_creation_month'] = df['trip_creation_time'].dt.month
df['trip_creation_day'] = df['trip_creation_time'].dt.day
df.head()
```

[21]:
```
       data         trip_creation_time  \
0  training 2018-09-20 02:35:36.476840
1  training 2018-09-20 02:35:36.476840
```

```
2  training 2018-09-20 02:35:36.476840
3  training 2018-09-20 02:35:36.476840
4  training 2018-09-20 02:35:36.476840

                                route_schedule_uuid route_type  \
0  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
1  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
2  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
3  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
4  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting

                trip_uuid source_center                 source_name  \
0  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
1  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
2  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
3  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
4  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)

  destination_center               destination_name  start_scan_to_end_scan  \
0       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
1       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
2       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
3       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0
4       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)                    86.0

   …  source_place   source_code destination_city  destination_place  \
0  …       VUNagar  DC (Gujarat)         Khambhat           MotvdDPP
1  …       VUNagar  DC (Gujarat)         Khambhat           MotvdDPP
2  …       VUNagar  DC (Gujarat)         Khambhat           MotvdDPP
3  …       VUNagar  DC (Gujarat)         Khambhat           MotvdDPP
4  …       VUNagar  DC (Gujarat)         Khambhat           MotvdDPP

   destination_code                   corridor     city_to_city  \
0       D (Gujarat)  DC (Gujarat)->D (Gujarat)  Anand->Khambhat
1       D (Gujarat)  DC (Gujarat)->D (Gujarat)  Anand->Khambhat
2       D (Gujarat)  DC (Gujarat)->D (Gujarat)  Anand->Khambhat
3       D (Gujarat)  DC (Gujarat)->D (Gujarat)  Anand->Khambhat
4       D (Gujarat)  DC (Gujarat)->D (Gujarat)  Anand->Khambhat

   trip_creation_year  trip_creation_month  trip_creation_day
0                2018                    9                 20
1                2018                    9                 20
2                2018                    9                 20
3                2018                    9                 20
4                2018                    9                 20

[5 rows x 39 columns]
```

## 0.6 4.In-depth Analysis

[22]:
```
# Creating the dictionaries for trip base data with suitable aggregation␣
↪functions
create_trip_dict={'trip_uuid':'first','od_time_diff_hour':
↪'sum','actual_distance_to_destination':'sum','actual_time':
↪'last','osrm_time':'sum','osrm_distance':'sum','segment_actual_time':
↪'sum','segment_osrm_time':'sum','segment_osrm_distance':'sum'}
```

[23]:
```
# Aggregating the numerical columns based on the trip_uuid
trip_base_agg=df.groupby('trip_uuid').agg(create_trip_dict)
trip_base_agg
```

[23]:
```
                                        trip_uuid  od_time_diff_hour  \
trip_uuid
trip-153671041653548748  trip-153671041653548748         728.008209
trip-153671042288605164  trip-153671042288605164          15.219568
trip-153671043369099517  trip-153671043369099517        4144.906395
trip-153671046011330457  trip-153671046011330457           3.349831
trip-153671052974046625  trip-153671052974046625          26.478517
…                                              …                 …
trip-153861095625827784  trip-153861095625827784          14.655464
trip-153861104386292051  trip-153861104386292051           2.019684
trip-153861106442901555  trip-153861106442901555          21.105993
trip-153861115439069069  trip-153861115439069069          22.006709
trip-153861118270144424  trip-153861118270144424          11.813586

                         actual_distance_to_destination  actual_time  \
trip_uuid
trip-153671041653548748                      8860.812105        732.0
trip-153671042288605164                       240.208306         47.0
trip-153671043369099517                     68163.502238        611.0
trip-153671046011330457                        28.529648         59.0
trip-153671052974046625                       239.007304        131.0
…                                                    …          …
trip-153861095625827784                       141.057373         49.0
trip-153861104386292051                        25.130640         21.0
trip-153861106442901555                        93.743842        190.0
trip-153861115439069069                       355.281673         90.0
trip-153861118270144424                       110.239116        233.0

                         osrm_time  osrm_distance  segment_actual_time  \
trip_uuid
trip-153671041653548748     7787.0     10577.7647               1548.0
trip-153671042288605164      210.0       269.4308                141.0
trip-153671043369099517    65768.0     89447.2488               3308.0
trip-153671046011330457       24.0        31.6475                 59.0
```

```
trip-153671052974046625      207.0       266.2914                340.0
…                            …            …                  …
trip-153861095625827784      148.0       162.9473                 82.0
trip-153861104386292051       19.0        26.5333                 21.0
trip-153861106442901555      134.0       162.8499                281.0
trip-153861115439069069      446.0       449.5383                258.0
trip-153861118270144424      106.0       127.8020                274.0

                         segment_osrm_time  segment_osrm_distance
trip_uuid
trip-153671041653548748             1008.0              1320.4733
trip-153671042288605164               65.0                84.1894
trip-153671043369099517             1941.0              2545.2678
trip-153671046011330457               16.0                19.8766
trip-153671052974046625              115.0               146.7919
…                                     …                   …
trip-153861095625827784               62.0                64.8551
trip-153861104386292051               11.0                16.0883
trip-153861106442901555               88.0               104.8866
trip-153861115439069069              221.0               223.5324
trip-153861118270144424               67.0                80.5787

[14817 rows x 9 columns]
```

[24]:
```
# creating a function for outlier detection
def outlier_removal(df_demo,column):
  Q1=df_demo[column].quantile(0.25)
  Q3=df_demo[column].quantile(0.75)
  IQR=Q3-Q1
  lower=Q1-(1.5*IQR)
  upper=Q3+(1.5*IQR)

  return df_demo[(df_demo[column]>=lower)&(df_demo[column]<=upper)]
```

[25]:
```
#removing outliers from the aggregated dataframe and storing it to a new␣
↪dataframe
column=['od_time_diff_hour','actual_time','osrm_time','actual_distance_to_destination','osrm_d
trip_base_agg_cpy=trip_base_agg.copy()
for col in column:
  trip_base_agg_cpy=outlier_removal(trip_base_agg_cpy,col)


# The hypothesis testing can be done on the newly created dataset␣
↪trip_base_agg_cpy where the outlier removal was done.
```

[26]:
```
# Applying one hot encoding for the categorical variables
df = pd.get_dummies(df,columns=['route_type'],drop_first=True)
```

[ ]:
```
df.head()
```

[ ]:
```
        data         trip_creation_time  \
0  training 2018-09-20 02:35:36.476840
1  training 2018-09-20 02:35:36.476840
2  training 2018-09-20 02:35:36.476840
3  training 2018-09-20 02:35:36.476840
4  training 2018-09-20 02:35:36.476840

                                route_schedule_uuid                trip_uuid  \
0  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…  trip-153741093647649320
1  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…  trip-153741093647649320
2  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…  trip-153741093647649320
3  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…  trip-153741093647649320
4  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…  trip-153741093647649320

  source_center                 source_name destination_center  \
0  IND388121AAA  Anand_VUNagar_DC (Gujarat)       IND388620AAB
1  IND388121AAA  Anand_VUNagar_DC (Gujarat)       IND388620AAB
2  IND388121AAA  Anand_VUNagar_DC (Gujarat)       IND388620AAB
3  IND388121AAA  Anand_VUNagar_DC (Gujarat)       IND388620AAB
4  IND388121AAA  Anand_VUNagar_DC (Gujarat)       IND388620AAB

                destination_name  start_scan_to_end_scan  is_cutoff  …  \
0  Khambhat_MotvdDPP_D (Gujarat)                    86.0       True  …
1  Khambhat_MotvdDPP_D (Gujarat)                    86.0       True  …
2  Khambhat_MotvdDPP_D (Gujarat)                    86.0       True  …
3  Khambhat_MotvdDPP_D (Gujarat)                    86.0       True  …
4  Khambhat_MotvdDPP_D (Gujarat)                    86.0      False  …

   source_place   source_code  destination_city  destination_place  \
0       VUNagar  DC (Gujarat)          Khambhat           MotvdDPP
1       VUNagar  DC (Gujarat)          Khambhat           MotvdDPP
2       VUNagar  DC (Gujarat)          Khambhat           MotvdDPP
3       VUNagar  DC (Gujarat)          Khambhat           MotvdDPP
4       VUNagar  DC (Gujarat)          Khambhat           MotvdDPP

   destination_code                   corridor  trip_creation_year  \
0       D (Gujarat)  DC (Gujarat)->D (Gujarat)                2018
1       D (Gujarat)  DC (Gujarat)->D (Gujarat)                2018
2       D (Gujarat)  DC (Gujarat)->D (Gujarat)                2018
3       D (Gujarat)  DC (Gujarat)->D (Gujarat)                2018
4       D (Gujarat)  DC (Gujarat)->D (Gujarat)                2018

   trip_creation_month  trip_creation_day  route_type_FTL
0                    9                 20           False
1                    9                 20           False
```

```
2                    9                    20           False
3                    9                    20           False
4                    9                    20           False

[5 rows x 38 columns]
```

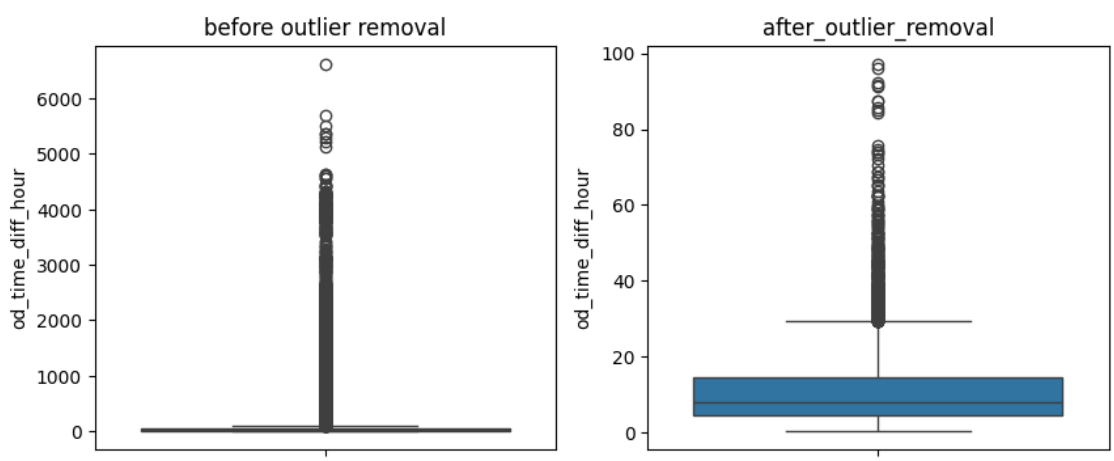
## 0.7 6. Visualization on outliers

[ ]:
```
# comparing the outliers before and after outlier removal from the trip base␣
↪aggregated dataset
```

[ ]:
```
#column=['od_time_diff_hour','actual_time','osrm_time','actual_distance_to_destination','osrm_
```

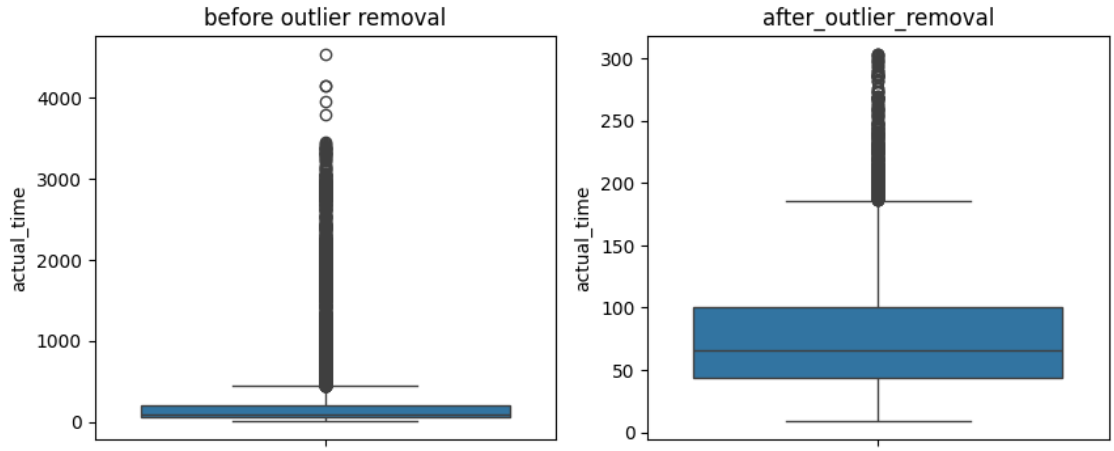
[ ]:
```
plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
sns.boxplot(trip_base_agg['od_time_diff_hour'])
plt.title('before outlier removal')
plt.subplot(1,2,2)
sns.boxplot(trip_base_agg_cpy['od_time_diff_hour'])
plt.title('after_outlier_removal')
```

[ ]:
```
Text(0.5, 1.0, 'after_outlier_removal')
```



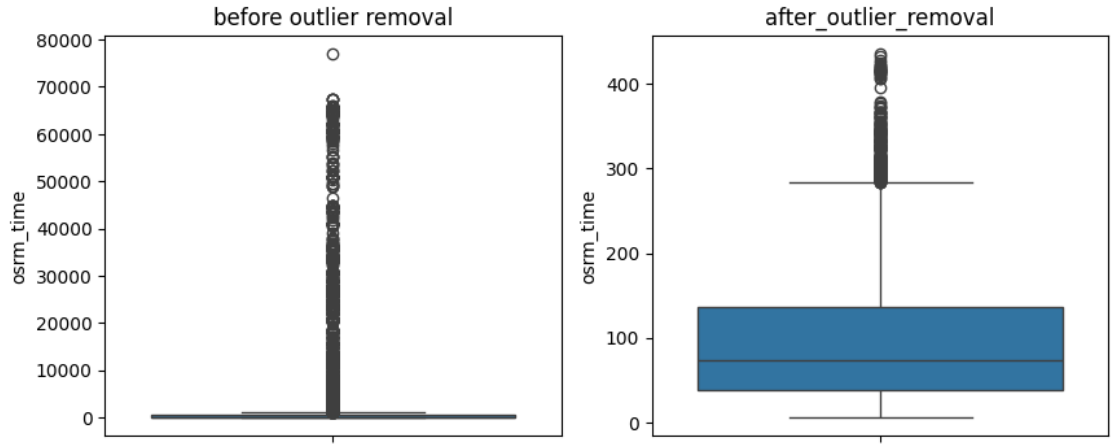

[ ]:
```
plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
sns.boxplot(trip_base_agg['actual_time'])
plt.title('before outlier removal')
plt.subplot(1,2,2)
sns.boxplot(trip_base_agg_cpy['actual_time'])
plt.title('after_outlier_removal')
```

[ ]: Text(0.5, 1.0, 'after_outlier_removal')




[ ]:
```
plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
sns.boxplot(trip_base_agg['osrm_time'])
plt.title('before outlier removal')
plt.subplot(1,2,2)
sns.boxplot(trip_base_agg_cpy['osrm_time'])
plt.title('after_outlier_removal')
```

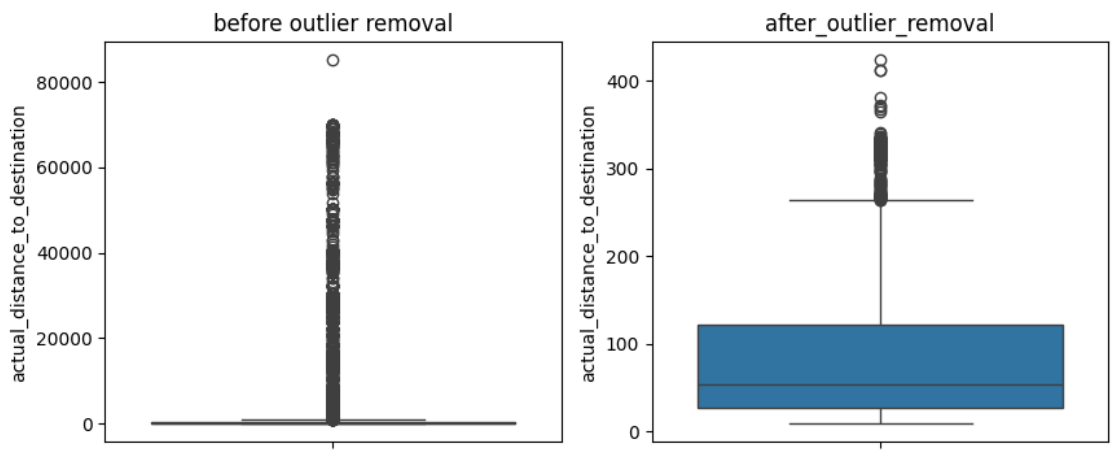
[ ]: Text(0.5, 1.0, 'after_outlier_removal')




[ ]:
```
plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
sns.boxplot(trip_base_agg['actual_distance_to_destination'])
```

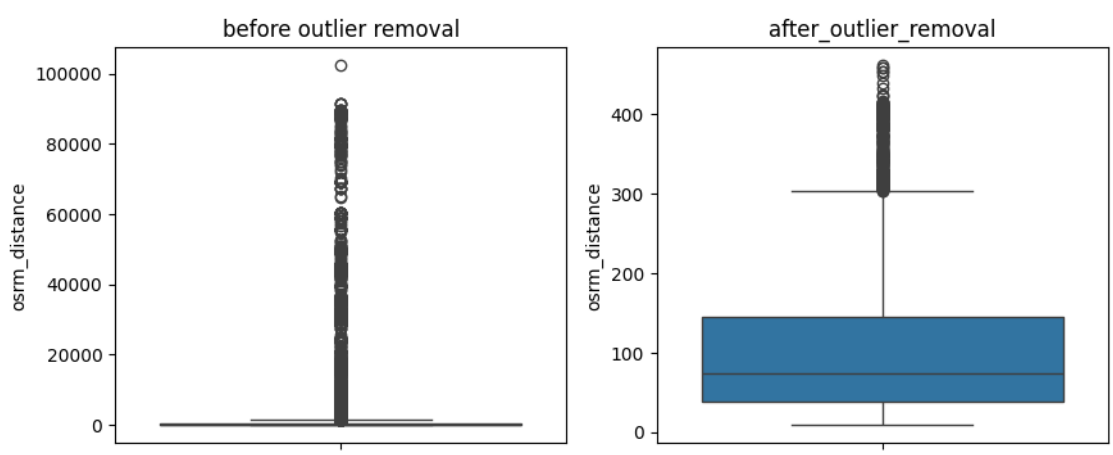
```
plt.title('before outlier removal')
plt.subplot(1,2,2)
sns.boxplot(trip_base_agg_cpy['actual_distance_to_destination'])
plt.title('after_outlier_removal')
```

[ ]: Text(0.5, 1.0, 'after_outlier_removal')



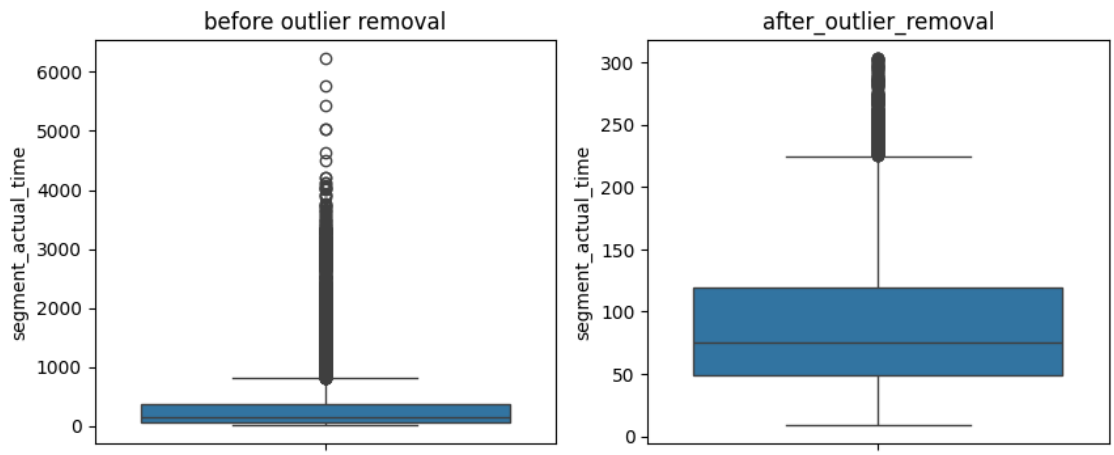

[ ]:
```
plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
sns.boxplot(trip_base_agg['osrm_distance'])
plt.title('before outlier removal')
plt.subplot(1,2,2)
sns.boxplot(trip_base_agg_cpy['osrm_distance'])
plt.title('after_outlier_removal')
```

[ ]: Text(0.5, 1.0, 'after_outlier_removal')



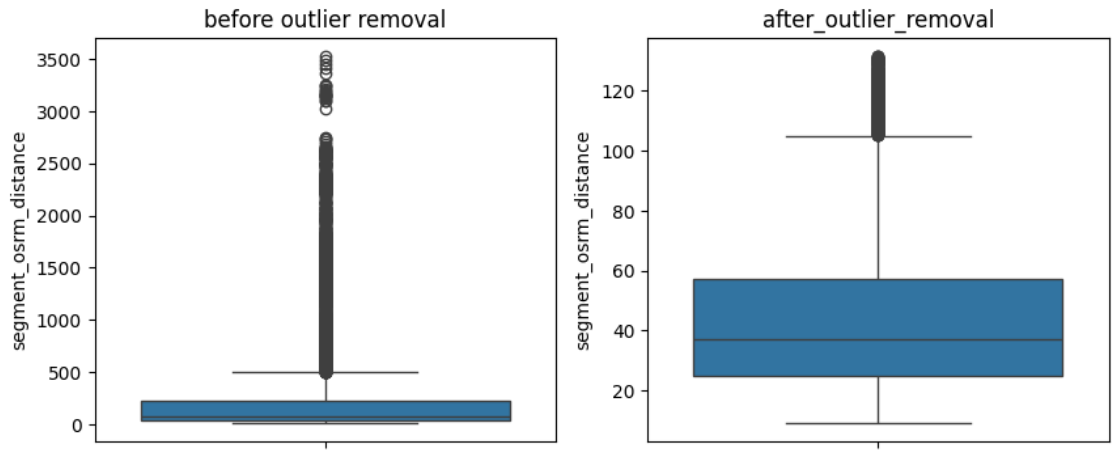
[ ]:
```
plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
sns.boxplot(trip_base_agg['segment_actual_time'])
plt.title('before outlier removal')
plt.subplot(1,2,2)
sns.boxplot(trip_base_agg_cpy['segment_actual_time'])
plt.title('after_outlier_removal')
```

[ ]: Text(0.5, 1.0, 'after_outlier_removal')



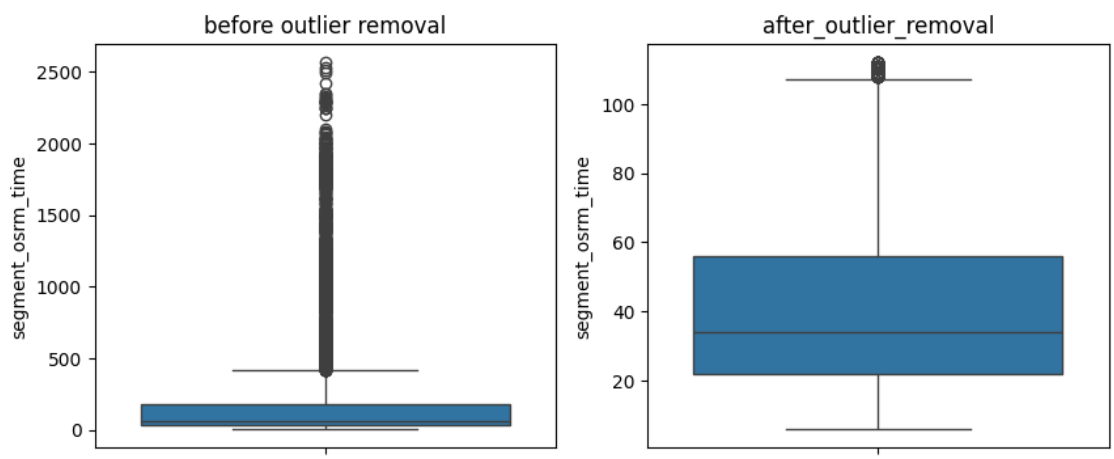

[ ]:
```
plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
sns.boxplot(trip_base_agg['segment_osrm_distance'])
plt.title('before outlier removal')
plt.subplot(1,2,2)
sns.boxplot(trip_base_agg_cpy['segment_osrm_distance'])
plt.title('after_outlier_removal')
```

[ ]: Text(0.5, 1.0, 'after_outlier_removal')




[ ]:
```
plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
sns.boxplot(trip_base_agg['segment_osrm_time'])
plt.title('before outlier removal')
plt.subplot(1,2,2)
sns.boxplot(trip_base_agg_cpy['segment_osrm_time'])
plt.title('after_outlier_removal')
```

[ ]:
```
Text(0.5, 1.0, 'after_outlier_removal')
```




[28]:
```
# Correlation and heat map for the numerical columns before and after the␣
↪outlier removal
df[column].corr()
```

[28]:

| | od_time_diff_hour | actual_time | osrm_time | \ |
|---|---|---|---|---|
| od_time_diff_hour | 1.000000 | 0.785936 | 0.785298 | |
| actual_time | 0.785936 | 1.000000 | 0.977998 | |
| osrm_time | 0.785298 | 0.977998 | 1.000000 | |
| actual_distance_to_destination | 0.785005 | 0.978659 | 0.995872 | |
| osrm_distance | 0.784137 | 0.979399 | 0.999119 | |
| segment_actual_time | 0.093302 | 0.124411 | 0.049892 | |
| segment_osrm_distance | 0.306980 | 0.242282 | 0.242282 | |
| segment_osrm_time | 0.219846 | 0.171465 | 0.177066 | |

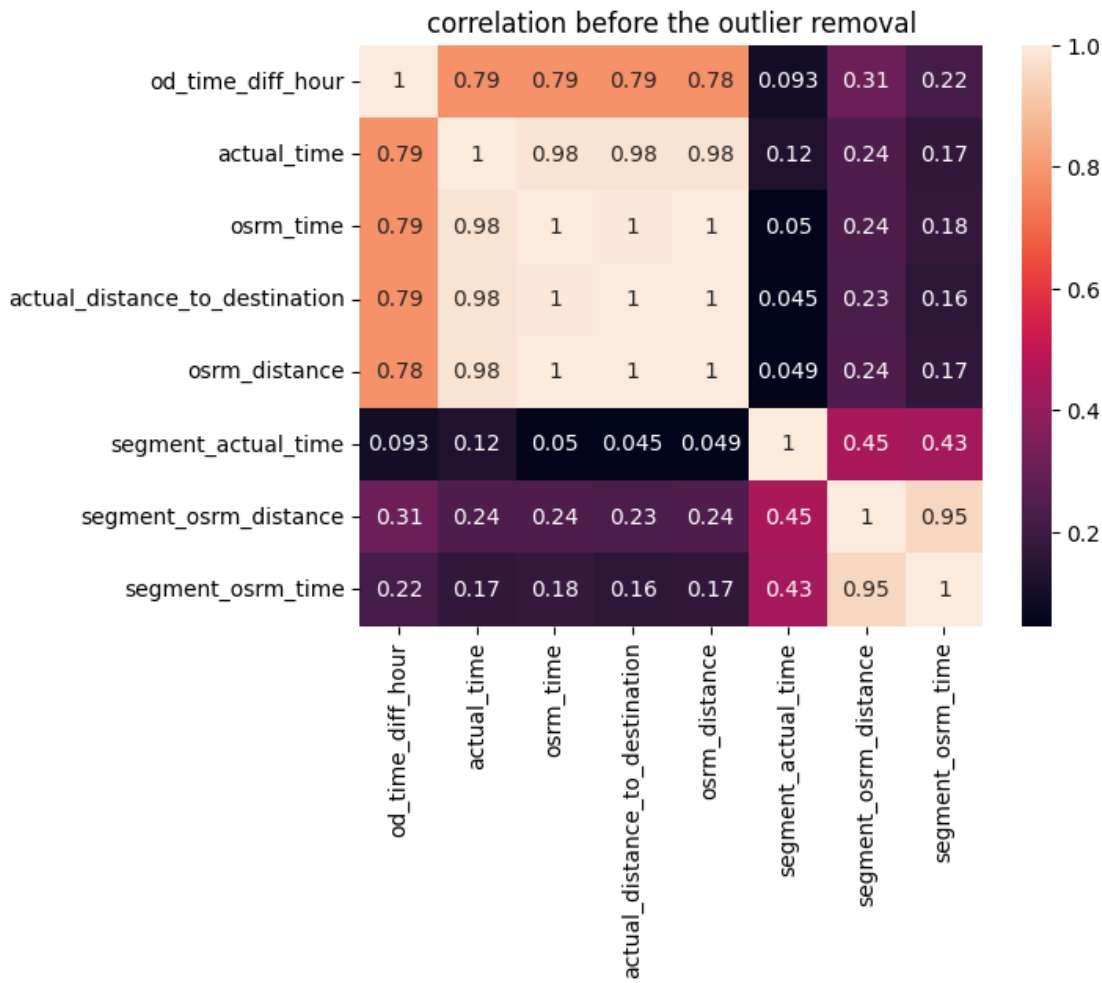
| | actual_distance_to_destination | osrm_distance | \ |
|---|---|---|---|
| od_time_diff_hour | 0.785005 | 0.784137 | |
| actual_time | 0.978659 | 0.979399 | |
| osrm_time | 0.995872 | 0.999119 | |
| actual_distance_to_destination | 1.000000 | 0.997149 | |
| osrm_distance | 0.997149 | 1.000000 | |
| segment_actual_time | 0.045241 | 0.048705 | |
| segment_osrm_distance | 0.232119 | 0.239669 | |
| segment_osrm_time | 0.158832 | 0.169151 | |

| | segment_actual_time | segment_osrm_distance | \ |
|---|---|---|---|
| od_time_diff_hour | 0.093302 | 0.306980 | |
| actual_time | 0.124411 | 0.242282 | |
| osrm_time | 0.049892 | 0.242282 | |
| actual_distance_to_destination | 0.045241 | 0.232119 | |
| osrm_distance | 0.048705 | 0.239669 | |
| segment_actual_time | 1.000000 | 0.448959 | |
| segment_osrm_distance | 0.448959 | 1.000000 | |
| segment_osrm_time | 0.433422 | 0.948523 | |

| | segment_osrm_time |
|---|---|
| od_time_diff_hour | 0.219846 |
| actual_time | 0.171465 |
| osrm_time | 0.177066 |
| actual_distance_to_destination | 0.158832 |
| osrm_distance | 0.169151 |
| segment_actual_time | 0.433422 |
| segment_osrm_distance | 0.948523 |
| segment_osrm_time | 1.000000 |

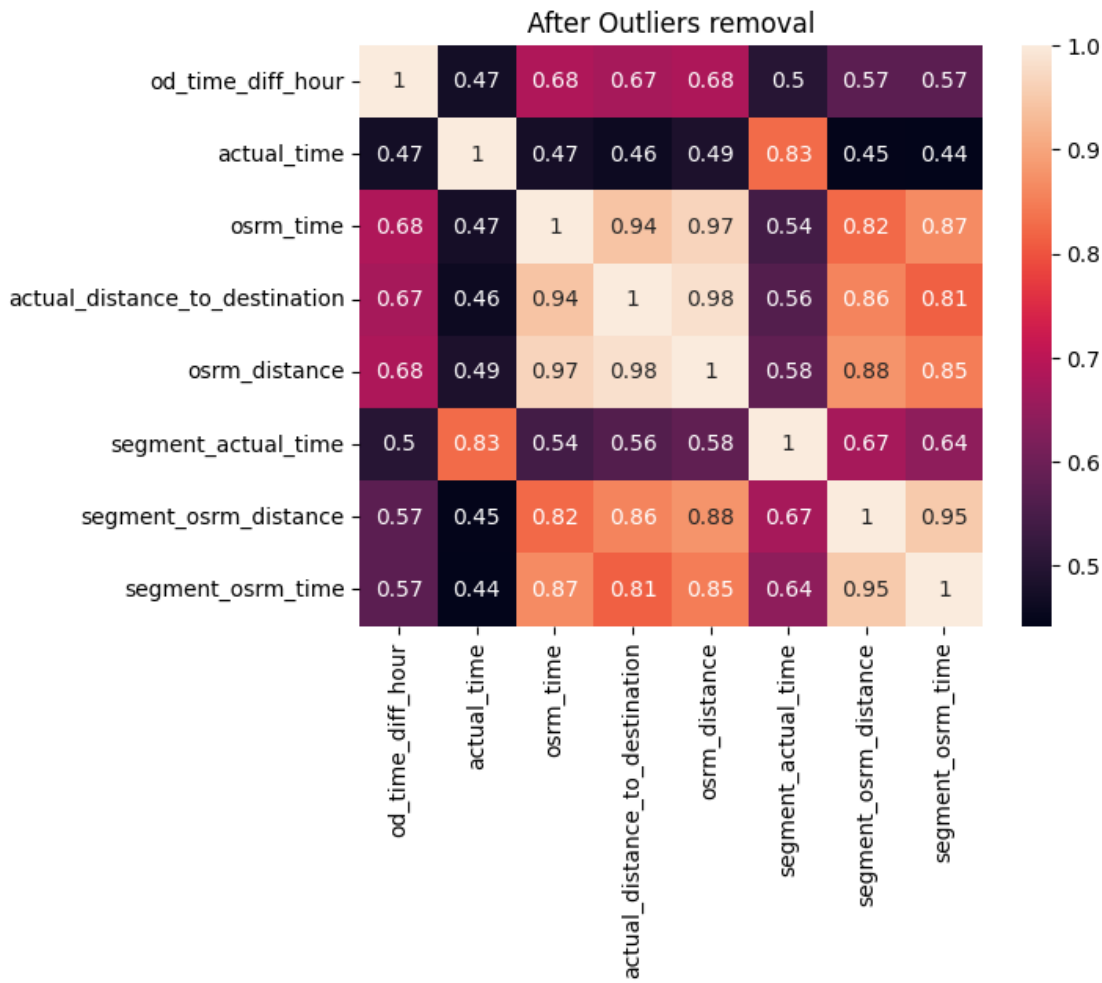
[36]:
```
sns.heatmap(df[column].corr(),annot=True)
plt.title('correlation before the outlier removal')
```

[36]: Text(0.5, 1.0, 'correlation before the outlier removal')




[37]:
```
sns.heatmap(trip_base_agg_cpy[column].corr(),annot=True)
plt.title('After Outliers removal')
```

[37]: Text(0.5, 1.0, 'After Outliers removal')




- Transportion time and distance are in strong positive correlation. The transportion time increases as the distance inreases
- The actual time and the predicted time are only weakly related.

## 0.8 5. Hypothesis Testing

1. actual_time aggregated value vs OSRM time aggregated value.

Is the actual time taken by the trip lesser than the predicted time?

- H0: There is no difference between the actual_time and osrm_time
- Ha: actual time different the predicted time

signicance level 5%

[68]:
```
ttest_rel(trip_base_agg_cpy['actual_time'],trip_base_agg_cpy['osrm_time'])
```

[68]:
```
TtestResult(statistic=np.float64(-26.350368447580458),
pvalue=np.float64(2.625235905137636e-147), df=np.int64(8729))
```

Conclusions: * pvalue<significance level * Null hypothesis is rejected * Actual and osrm predicted time are different

**2.** actual_time aggregated value and segment actual time aggregated value.

Is the actual time aggregated and segment actual time aggregated values are same or not?

- H0: Actual time aggregated and segment actual time aggregated values are the same
- Ha : Actual time aggregated and segment actual time aggregated values are not the same Significance level 5%

[ ]:
```
ttest_rel(trip_base_agg_cpy['actual_time'],trip_base_agg_cpy['segment_actual_time'])
```

[ ]:
```
TtestResult(statistic=np.float64(-35.66091843880159),
pvalue=np.float64(3.85097858257603e-260), df=np.int64(8729))
```

Conclusions: * pvalue<significance level * Null hypothesis is rejected * Actual and osrm segment actual time aggregated are different

**3.** OSRMdistance aggregated value and segment OSRM distance aggregated value.

Is ther any significant difference between osrm distance aggregated value and segment osrm distance aggregated value

- H0: osrm distance aggregated value and segment osrm distance aggregated value same
- Ha: osrm distance aggregated value and segment osrm distance aggregated value are different

[ ]:
```
ttest_rel(trip_base_agg_cpy['osrm_distance'],trip_base_agg_cpy['segment_osrm_distance'])
```

[ ]:
```
TtestResult(statistic=np.float64(93.28087341975727), pvalue=np.float64(0.0),
df=np.int64(8729))
```

Conclusions: * pvalue<significance level * Null hypothesis is rejected * osrm distance and segment osrm predicted distance are different

**4.** OSRMtimeaggregated value and segment OSRM time aggregated value.

- Ha: osrm time aggregated and segment osrm time aggregated are the same
- Ha: osrm time aggregated and segment osrm time aggregated are not the same

significance level 5%

[ ]:
```
ttest_rel(trip_base_agg_cpy['osrm_time'],trip_base_agg_cpy['segment_osrm_time'])
```

[ ]:
```
TtestResult(statistic=np.float64(94.59804615974909), pvalue=np.float64(0.0),
df=np.int64(8729))
```

Conclusions: * pvalue<significance level * Null hypothesis is rejected * osrm time and segment aggregated osrm time are different

## 0.9 7. Feature Scaling

[ ]:
```
scaler=StandardScaler()
column=['od_time_diff_hour','actual_time','osrm_time','actual_distance_to_destination','osrm_d
for col in column:
  trip_base_agg_cpy[col] = scaler.fit_transform(trip_base_agg_cpy[[col]])
```

- Applied feature scaling to the numerical columns in the aggregated data sate based on the trip_uuid

## 0.10 8. Business Recommendations

- Delivery time:
  - The actual time taken during transport and predicted time are not the same. So the model for predicting the time can be made better for more accurate prediction
  - Difference between Actual time and predicted time is positive which tells, delivery of orders happens delayed than the expected time.
  - Delay in the actual transport time can be reduced by reducing the expected delays and choosing the less traffic routes *Route type:
  - As the full truck load type takes the lowest time when compared to carting method, more transportation can be done through this method
  - More orders can be shipped to the destination hub through busiest routes where more trucks are available
  - Tracking alerts can be given for better monitoring
- Source and destination hubs can be expanded to more tier2 and tier 3 cities where the demand is higher
- HB (Haryana)->H (Karnataka) is the busiest route. Increasing truck allocation in this route can reduce the delays.
- Average time for tranportation is more for the corridor Punjab to Karnataka. This is longer route. Intermediate hubs can be introduced for longer routes to reduce the transporation time. Better routes (like express highways) can be selected for the transportation.
- loading and unloading time should be optimised

[ ]:
```

```